



CDH10 (CHR5)

Gene Report

Prepared for the CHDI Foundation
by Rancho BioSciences

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1. General Gene Information

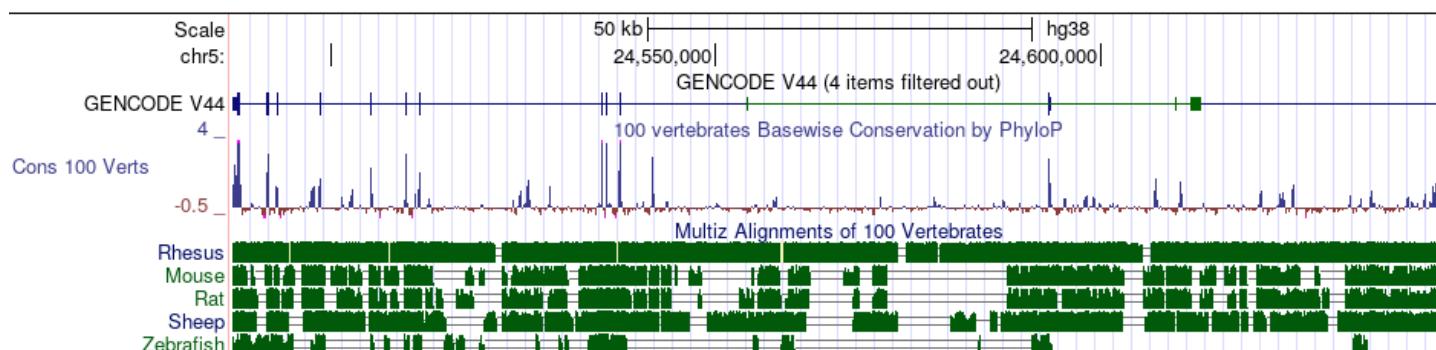
a. Gene Aliases

	Human gene	Mouse gene	Rat gene
Entrez Gene ID	1008	320873	29181
Gene Symbol	CDH10	Cdh10	Cdh10
Gene Name	cadherin 10	cadherin 10	cadherin 10
Ensembl ID	ENSG00000040731	ENSMUSG00000022321	ENSRNOG00000009771
Uniprot ID	Q9Y6N8	P70408	F1LR98
Synonyms/Aliases	Cadherin-10, T2-cadherin, Cadherin 10 Type 2	A830016G23Rik, C030003B10Rik, C030011H18Rik, T2-cadherin	None

b. Conservation of gene across species and conservation of region under peak in other genomes

The UCSC Genome browser hosts the complete human genome alongside different versions of human genome assembly and includes a large collection of vertebrate and model organism assemblies and annotations. This tool provides a graphical interface for viewing, analyzing, and downloading data. A query to assess conservation revealed available information for 4 transcript variants of CDH10. UCSC Genome Browser's "conservation" tracks of transcript variant 1 are shown below.

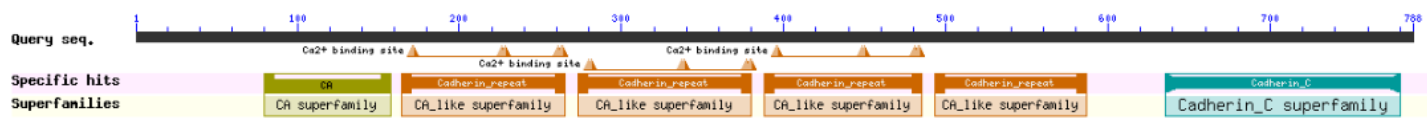
[CDH10 \(ENST00000264463.8\) - chr5:24487100-24644978](#) - Homo sapiens cadherin 10 (CDH10), transcript variant 1, mRNA. (from RefSeq NM_006727)



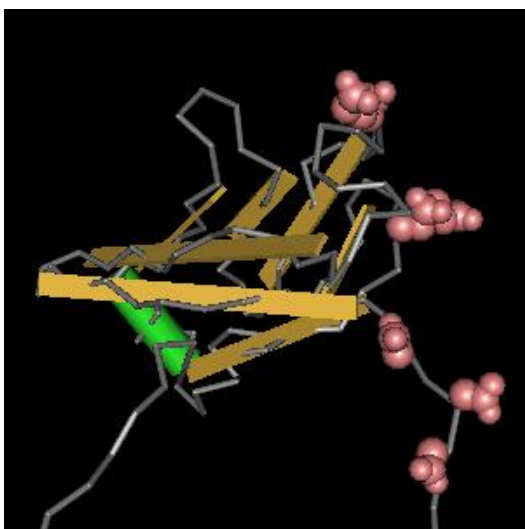
c. Known structure

The NCBI Structure Group and Conserved Domain Database (CDD) are protein annotation resources utilizing multiple sequence alignment models for identification of conserved domains in protein sequences. The RCSB Protein Data Bank (PDB) is another open resource for assessing protein structure, providing access to 3D structure data for large biological molecules (proteins, DNA, and RNA). Below is a subset of the most relevant domains and structures, with links to additional information.

Conserved domains for CDH10:



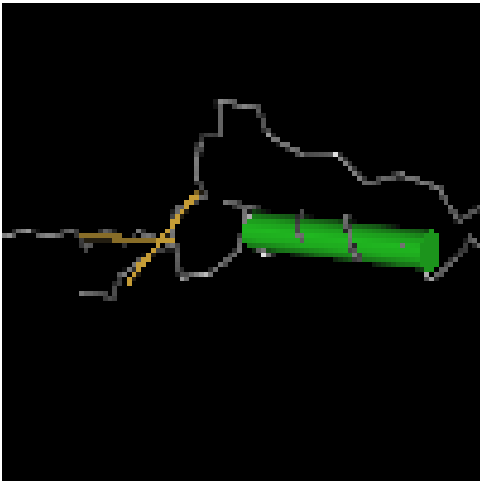
Ca2+ binding site [ion binding site] ([NCBI CDD Link](#)):



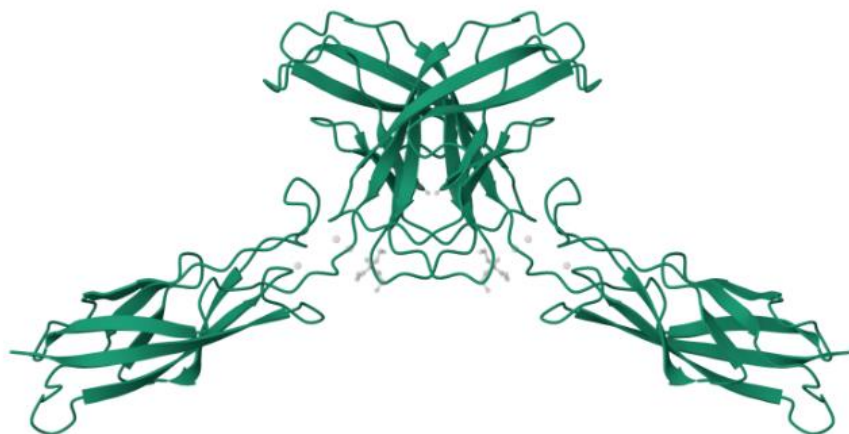
CD11304: Cadherin_repeat: Cadherins are glycoproteins involved in Ca2+-mediated cell-cell adhesion. The cadherin repeat domains occur as tandem repeats in the extracellular regions, which are thought to mediate cell-cell contact when bound to calcium. They play numerous roles in cell fate, signalling, proliferation, differentiation, and migration; members include E-, N-, P-, T-, VE-, CNR-, proto-, and FAT-family cadherin, desmocollin, and desmoglein, a large variety of domain architectures with varying repeat copy numbers. Cadherin-repeat containing proteins exist as monomers, homodimers, or heterodimers. ([NCBI CDD Link](#))



Cadherin_C: Cadherin cytoplasmic region: Cadherins are vital in cell-cell adhesion during tissue differentiation. Cadherins are linked to the cytoskeleton by catenins. Catenins bind to the cytoplasmic tail of the cadherin. Cadherins cluster to form foci of homophilic binding units. A key determinant to the strength of the binding that is mediated by cadherins is the juxtamembrane region of the cadherin. This region induces clustering and binds to the protein p120ctn ([NCBI CDD Link](#)).

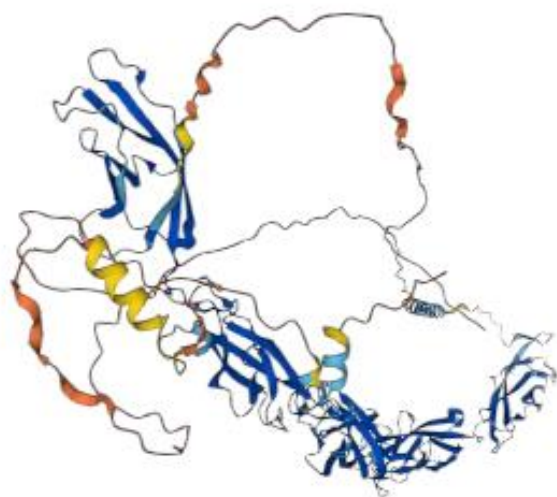


3D structures from PDB: Mouse CDH10 protein expressed in E. Coli ([PDB link](#))



d. AlphaFold protein structure prediction

Human CDH10: [AlphaFold Protein Structure Link](#)



Model Confidence ⓘ

- Very high (pLDDT > 90)
- High (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very low (pLDDT < 50)

AlphaFold produces a per-residue model confidence score (pLDDT) between 0 and 100. Some regions below 50 pLDDT may be unstructured in isolation.

Rat Cdh10: [AlphaFold Protein Structure Link](#)

Mouse Cdh10: [AlphaFold Protein Structure Link](#)

e. Homologues in model animals

A query of the human CDH10 gene in NCBI found 210 orthologs genes in tetrapods. Orthologs of CDH10 were found in tetrapods including house mouse, chicken, cattle, monkey, chimpanzee, dog, frog, pigeon, naked mole-rat, bat, etc. NCBI Orthologs for Human CDH10 gene and protein ([Link](#)) are shown below and additional orthologs are available in OrthoDB ([Link](#)).

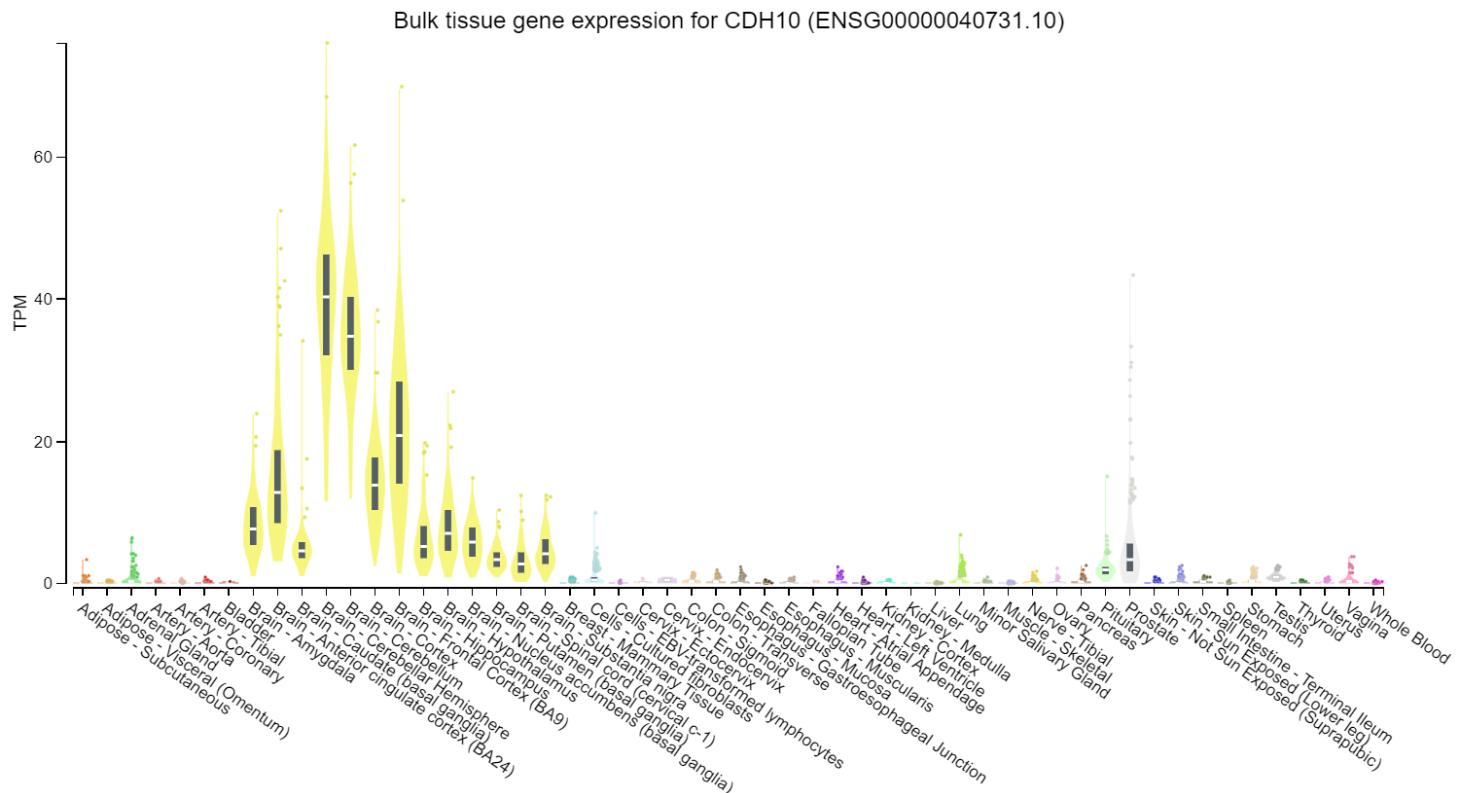
Species	Gene	Architecture	aa
☐ <i>Mus musculus</i> house mouse	Cdh10 cadherin 10		788
☐ <i>Gallus gallus</i> chicken	CDH10 cadherin 10		789
☐ <i>Bos taurus</i> cattle	CDH10 cadherin 10		788
☐ <i>Macaca mulatta</i> Rhesus monkey	CDH10 cadherin 10		788
☐ <i>Pan troglodytes</i> chimpanzee	CDH10 cadherin 10		788
☐ <i>Canis lupus familiaris</i> dog	CDH10 cadherin 10		788
☐ <i>Xenopus tropicalis</i> tropical clawed frog	cdh10 cadherin 10, type 2 (T2-cadherin)		787
☐ <i>Columba livia</i> rock pigeon	CDH10 cadherin 10		789
☐ <i>Macaca fascicularis</i> crab-eating macaque	CDH10 cadherin 10		788
☐ <i>Heterocephalus glaber</i> naked mole-rat	Cdh10 cadherin 10		788
☐ <i>Bos mutus</i> wild yak	CDH10 cadherin 10		788
☐ <i>Myotis brandtii</i> Brandts bat	CDH10 cadherin 10		788
☐ <i>Callithrix jacchus</i> white-tufted-ear marmoset	CDH10 cadherin 10		788
☐ <i>Trachemys scripta elegans</i>	CDH10 cadherin 10		768
☐ <i>Nannospalax galili</i> Upper Galilee mountains blind mole rat	Cdh10 cadherin 10		788
☐ <i>Caretta caretta</i> Loggerhead turtle	CDH10 cadherin 10		789
☐ <i>Chelonia mydas</i> Green sea turtle	CDH10 cadherin 10		789
☐ <i>Chrysemys picta</i> Painted turtle	CDH10 cadherin 10		789
☐ <i>Alligator mississippiensis</i> American alligator	CDH10 cadherin 10		789
☐ <i>Crocodylus porosus</i>	CDH10		789

2. Expression pattern by cell and tissue

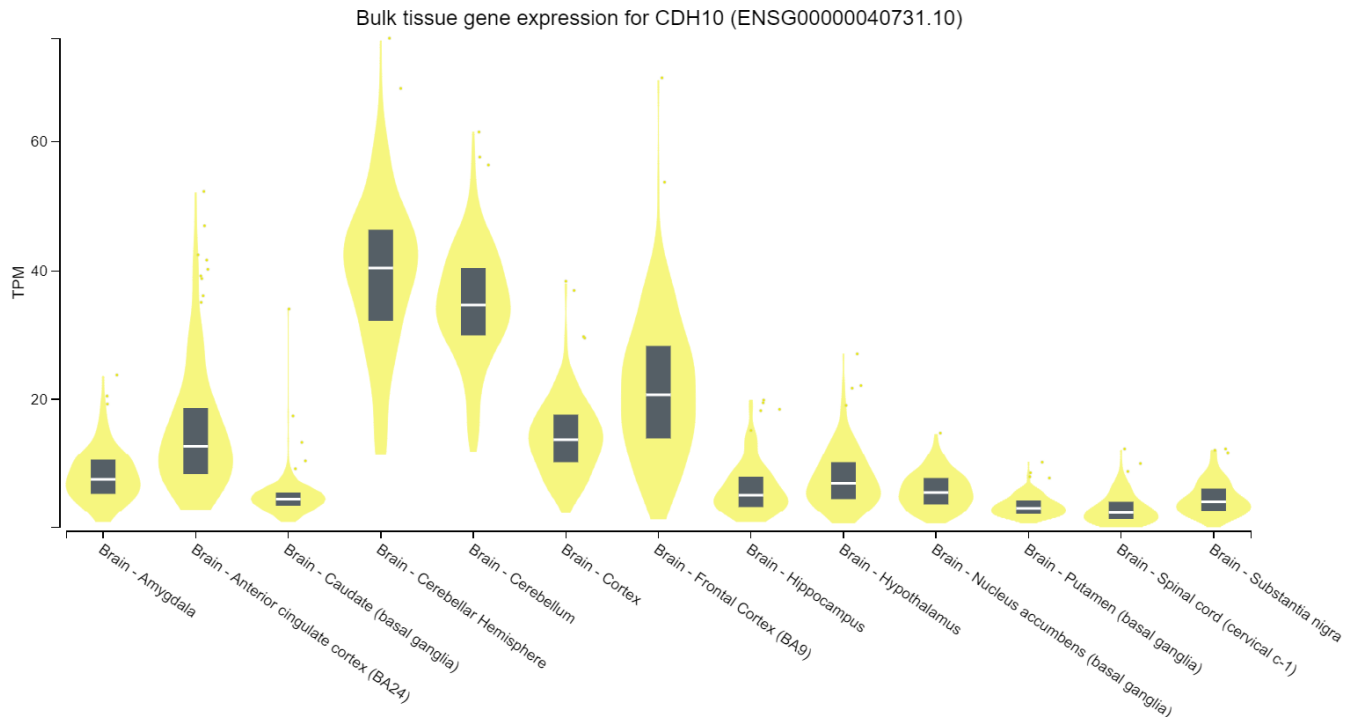
a. Tissue-specific information

The Genotype-Tissue Expression ([GTEx](#)) project is a comprehensive public resource that provides tissue-specific gene expression and regulation data, including RNA-Seq, eQTLs, and histology images. The resource contains information collected from 53 non-diseased tissue sites across nearly 1000 individuals, utilizing primarily molecular assays such as whole genome/exome sequencing and RNA-Seq. Below is information pertaining to CDH10 expression across all tissues and specifically in the brain.

[GTEx](#): CDH10 expression in all tissues

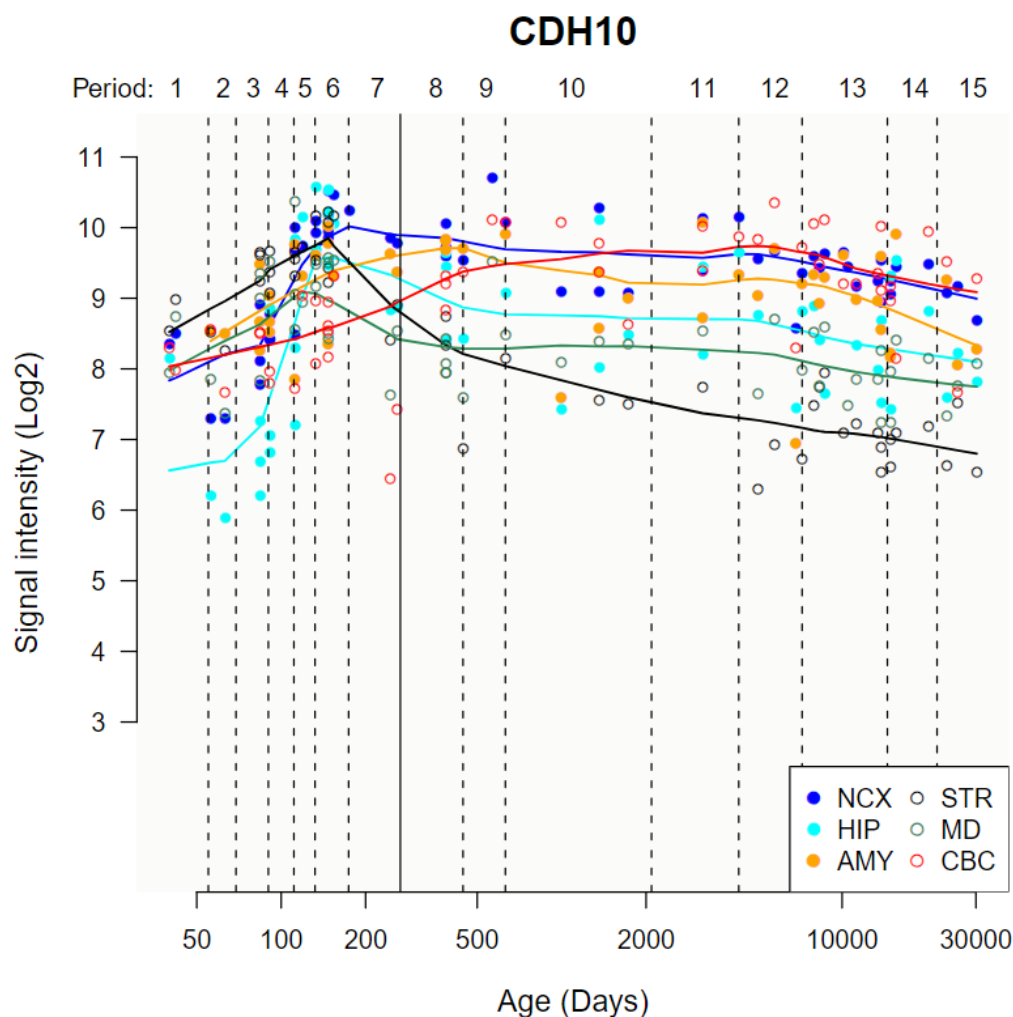


GTEX: CDH10 expression in brain



The Human Brain Transcriptome ([HBT](#)) is a public database hosted by the Department of Neurobiology at Yale University School of Medicine. It contains transcriptome data and associated metadata from over 1,340 tissue samples, specifically focusing on the developing and adult human brain.

CDH10 gene expression ([Link](#)) along entire development and adulthood in the cerebellar cortex (CBC), mediodorsal nucleus of the thalamus (MD), striatum (STR), amygdala (AMY), hippocampus (HIP) and 11 areas of neocortex (NCX) is shown below.



b. Barres Lab RNA-Seq brain specific expression data

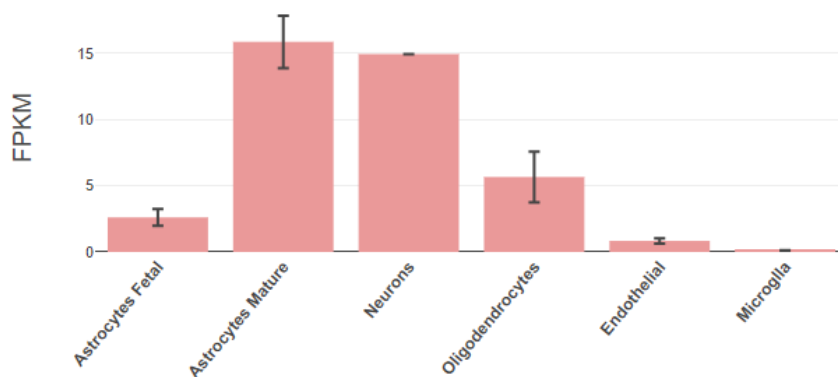
Expression data from Allen Brain Atlas and Barres Lab RNA-Seq were downloaded, and average expression levels were calculated. The CDH10 expression results are presented in the table below.

Average Human Brain Agilent Expression	Average Human Brain RNA-Seq Expression (TPM)	Average Mouse Brain RNA-Seq Expression (TPM)	Human Brain Expression Category	Mouse Brain Expression Category
7.01152731	48.69530495	44.97165541	medium	high

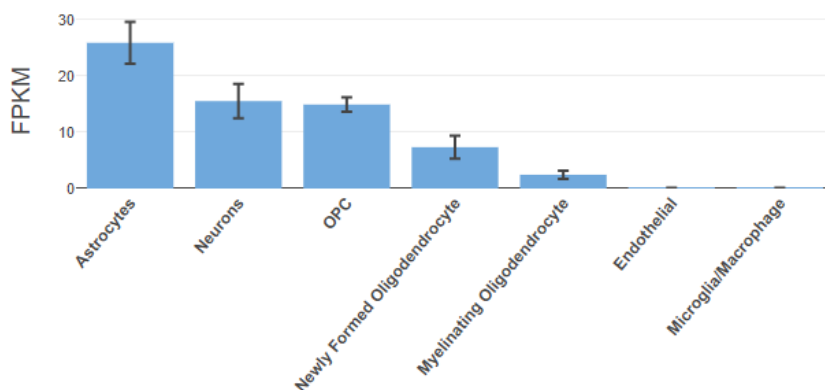
Additionally, relative CDH10 RNA expression in various cell types isolated from human and mouse brain as measured by RNA-Seq is shown here, where FPKM is Fragments Per Kilobase Million, a normalization unit commonly utilized in this types of assay ([Barres Lab Brain RNA-Seq](#)).

Brain RNA-Seq generated by the Ben Barres lab ([Link](#))

CDH10 - Homo sapiens



Cdh10 - Mus musculus

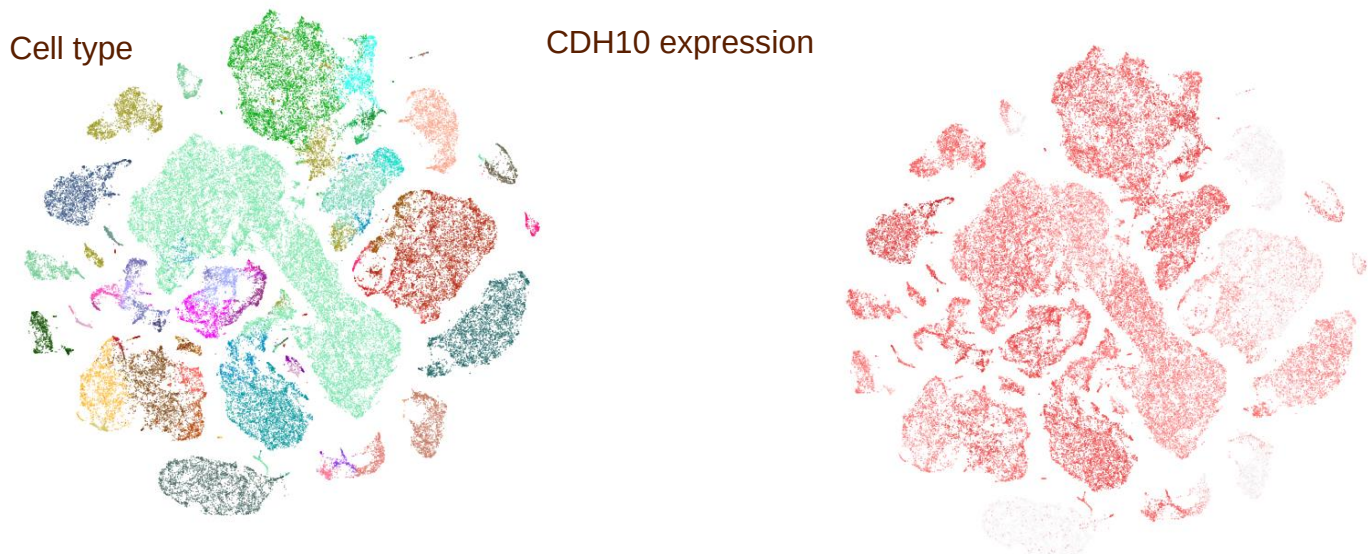


c. snRNA-Seq gene expression in cell type database

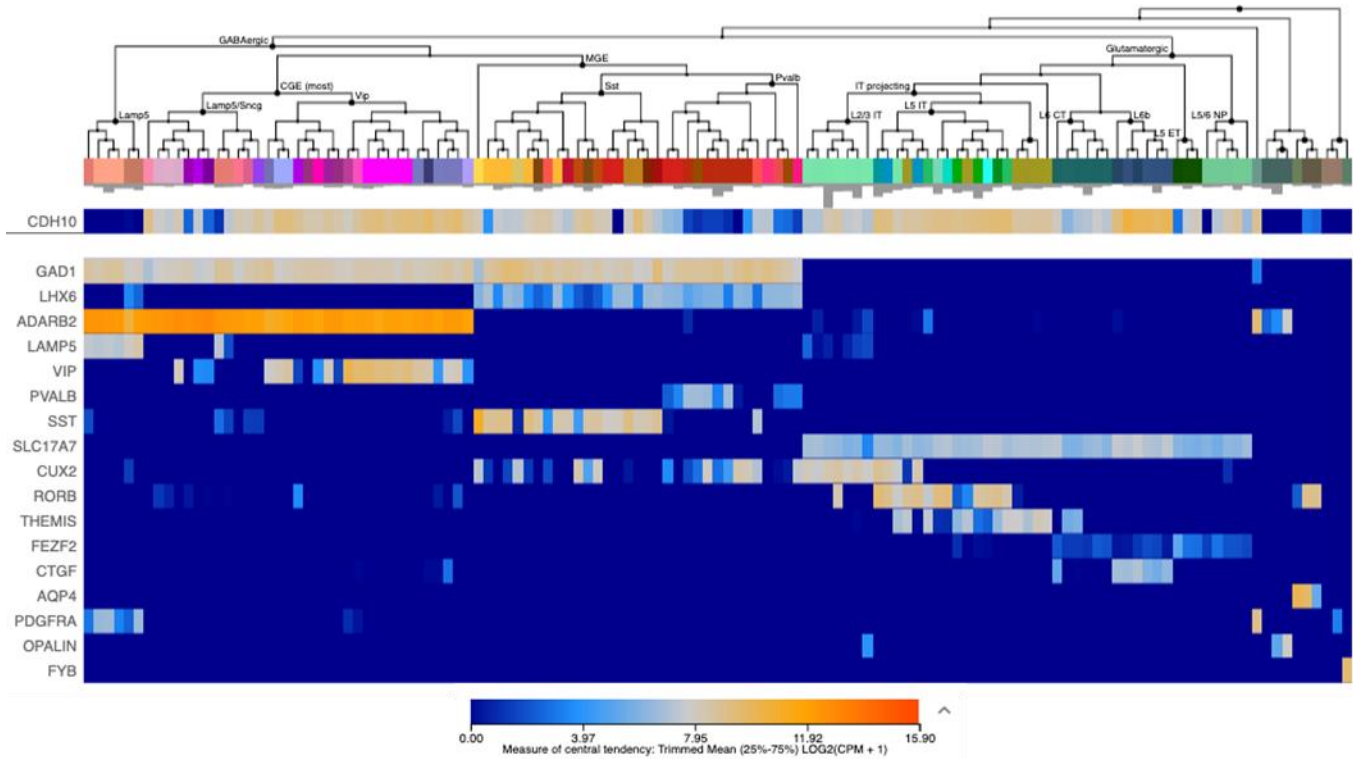
1. Single cell (or single nucleus) RNA sequencing (RNA-Seq) is a scalable approach to provide genome-wide expression profiles for thousands of cells. This data set includes single cell and nuclear transcriptomic profiles, assayed from human and mouse brain regions. Anatomical specificity is achieved by microdissecting tissue from defined brain areas, such as cortical layers or cell groups in LGd. Individual layers of cortex were dissected from tissues covering the middle temporal gyrus

(MTG), anterior cingulate gyrus (CgGr), primary visual cortex (V1C), primary motor cortex (M1C), primary somatosensory cortex (S1C) and primary auditory cortex (A1C) derived from human brain, and nuclei were dissociated and sorted using the neuronal marker NeuN. Nuclei were sampled from postmortem and neurosurgical (MTG only) donor brains and expression was profiled with SMART-Seq v4 or 10x Genomics Chromium Single Cell 3' v3 RNA-sequencing. Analysis for gene of interest in the [Cell Types Database](#) for selected datasets were shown below:

Human-M1-10x motor cortex dataset from Cell Type Database: (Single cell RNA-seq data): For the Human-M1-10x motor cortex dataset, the left UMAP plot was overlaid and colored by cell sub-type. The large light green cluster in the middle represents the excitatory neurons from L2. The UMAP in the right plot shows the expression of CDH10 overlaid onto the UMAP. Each point represents a cell, and the darker the red points appear, the greater the expression of CDH10. The expression of CDH10 is lower in IN-Pvalb interneurons and higher in Vip, Sst, L5IT, and L6b.



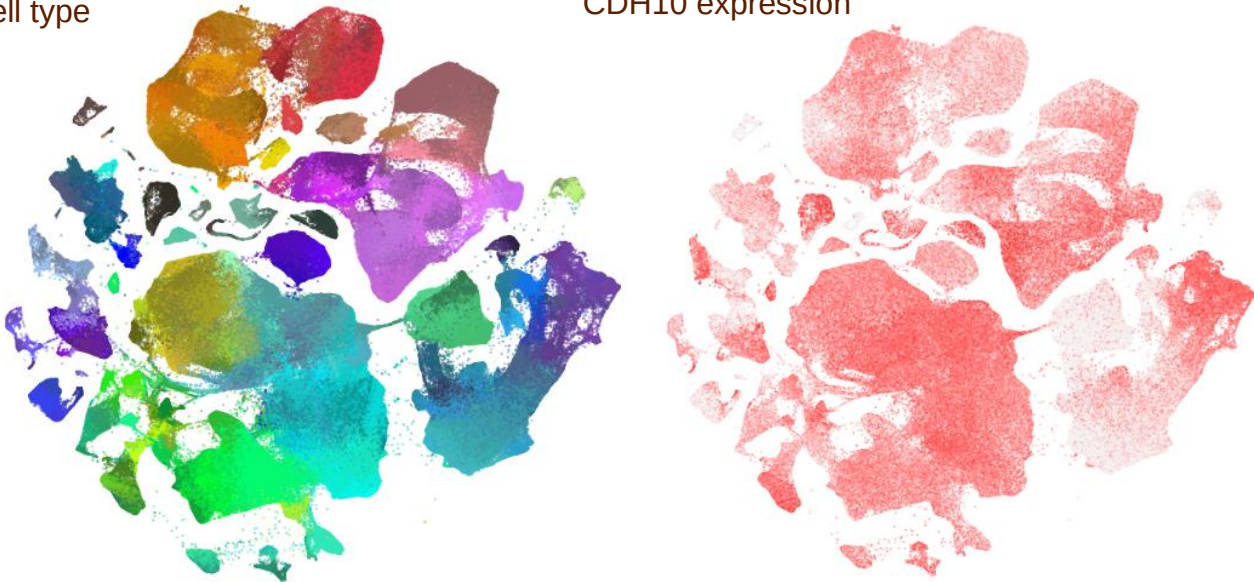
Human-M1-10x motor cortex dataset: [gene expression heatmap](#) (Single cell RNA-seq data): The heatmap below shows the expression of CDH10 (first row) across all cell types in the Human-M1-10x motor cortex dataset as compared to several other genes. While GAD1 and SLC17A7 primarily underly cell type specific dendrogram construction, CDH10 is highly expressed in both GABAergic and Glutamatergic neurons (specifically Vip GABAergic interneurons and Sst interneurons, L5IT, and L6b Glutamatergic neurons).



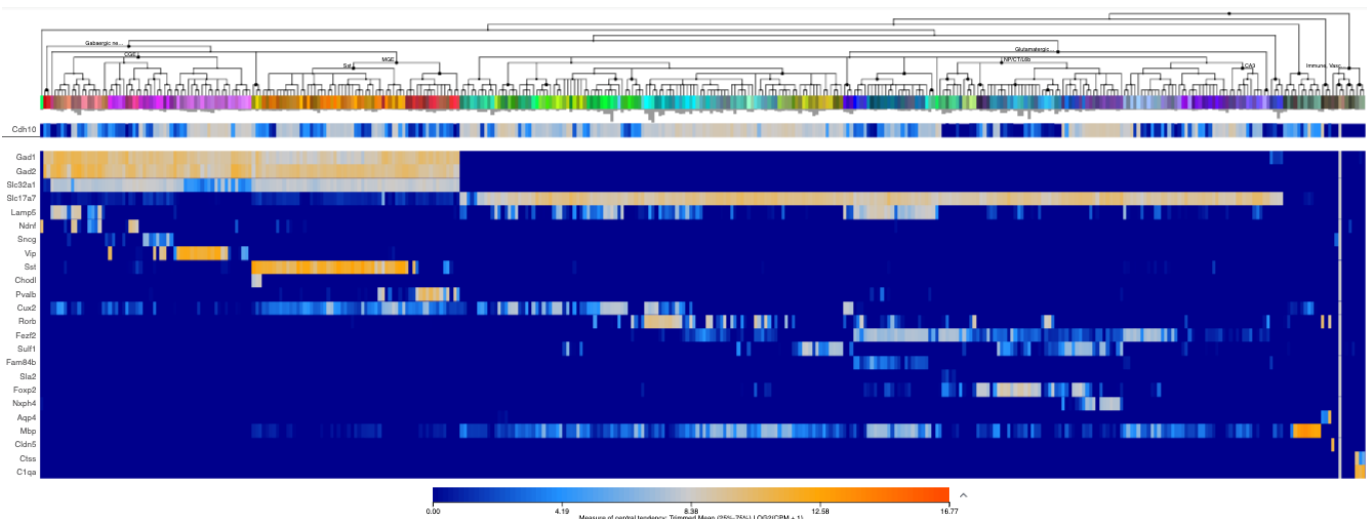
Mouse - WHOLE CORTEX & HIPPOCAMPUS - 10X GENOMICS (2020) dataset: For the Mouse – Whole Cortex and Hippocampus 10x dataset, the left UMAP plot was overlaid and colored by cell sub-type. The large bright red cluster at the top represents IN-Pvalb interneurons. The UMAP in the right plot shows the expression of Cdh10 overlaid onto the UMAP. Each point represents a cell, and the darker the red points appear, the greater the expression of Cdh10. As is clear from the UMAP, the expression of Cdh10 is higher in L6 IT, L5 IT and L4 IT in the cortex compared to some other cell types. There are other cell types that show high expression such as Astrocytes.

Cell type

CDH10 expression



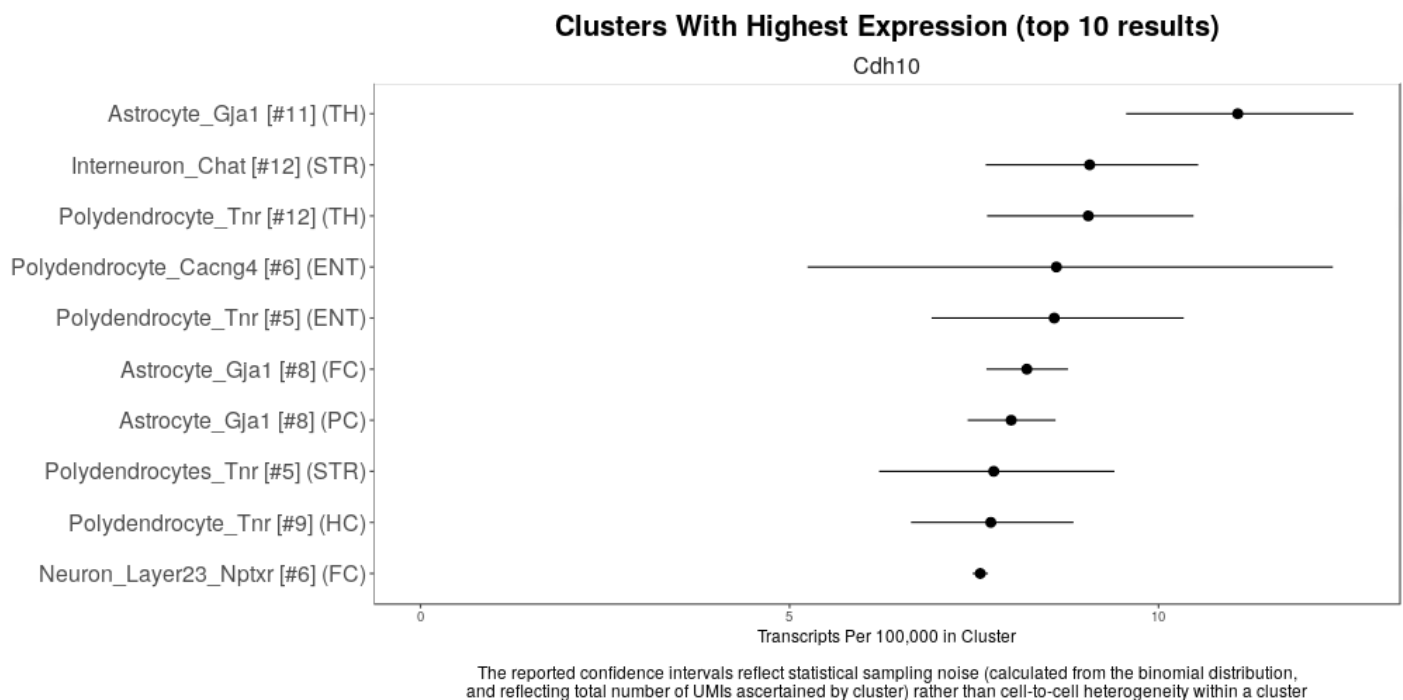
Mouse - WHOLE CORTEX & HIPPOCAMPUS - 10X GENOMICS (2020) dataset: gene expression heatmap (Single cell RNA-seq data): For the Mouse – Whole Cortex and Hippocampus 10x dataset, the expression of Cdh10 (first row) is shown below across all cell types as compared to other genes. While Gad1, Gad2 and Slc17a7 primarily underly cell type specific dendrogram construction, Cdh10 expression is high across several cell types in both GABAergic and Glutamatergic neurons.



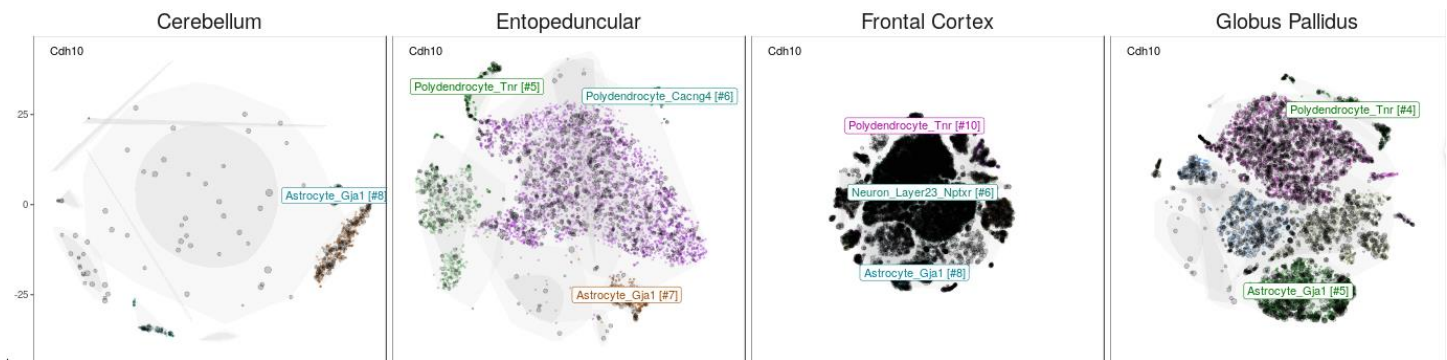
2. The mammalian brain is composed of diverse, specialized cell populations. To systematically ascertain and learn from these cellular specializations, Drop-seq was used to profile RNA expression in 690,000 individual cells sampled from 9 regions of the adult mouse brain. The value of DropViz is

that it contains data from the entire mouse brain. Authors identified 565 transcriptionally distinct groups of cells using computational approaches developed to distinguish biological from technical signals. Cross-region analysis of these 565 cell populations revealed features of brain organization, including a gene-expression module for synthesizing axonal and presynaptic components, patterns in the co-deployment of voltage-gated ion channels, functional distinctions among the cells of the vasculature and specialization of glutamatergic neurons across cortical regions. Systematic neuronal classifications for two complex basal ganglia nuclei and the striatum revealed a rare population of spiny projection neurons. This adult mouse brain cell atlas, accessible through interactive online software ([DropViz](#)), serves as a reference for development, disease, and evolution.

Levels by cluster for CDH10: The greatest expression of Cdh10 is found in the astrocytes cluster 11; however, the distribution of transcripts per 100,000 in the cluster overlaps with the majority of the other cell types with high expression of Cdh10.



tSNE analysis of global clusters for Cdh10: The tSNE plot shows expression of Cdh10 expression overlaid onto UMAPs by brain region. The greatest expression of Cdh10 is found in the oligodendrocytes of the Entopeduncular and Globus Pallidus brain regions. There is very little expression of Cdh10 found in the Cerebellum or the Frontal Cortex.



3. Perturbations in GEO that alter the gene

a. GEO Profiles search focusing on brain and/or neurons

The Gene Expression Omnibus (GEO) repository is a public database that stores MIAME (Minimum Information About a Microarray Experiment)- and MINSEQE (Minimum Information About a Next-generation Sequencing Experiment)-compliant gene expression data. GEO enables the search for pre-computed profiles, or a specific profile based on gene annotation. A search of the [GEO Profiles](#) database for datasets containing CDH10 yielded 5344 datasets. Filtering for up/down genes decreased the number of datasets to 87.

The following GEO profiles were selected as potentially relevant:

1. [Cdh10 - Fetal brain response to maternal immune activation induced by lipopolysaccharide](#)

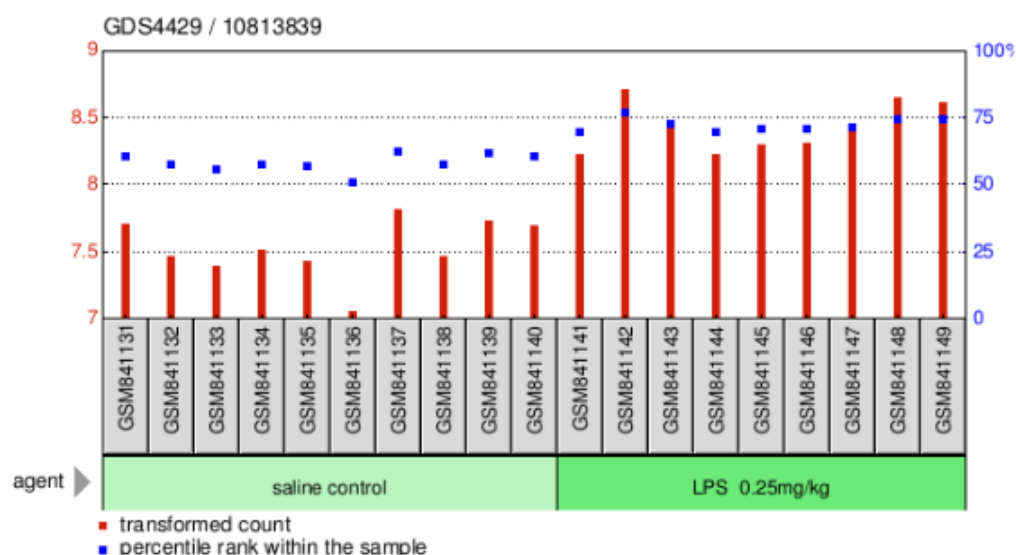
Annotation: Cdh10, cadherin 10

Organism: Rattus norvegicus

Reporter: GPL6247, 10813839 (ID_REF), GDS4429, AF468011, chr2:69561671-69696004 (SPOT ID)

DataSet type: Expression profiling by array, transformed count, 19 samples

ID: 92166261



2. Cdh10 - Survival motor neuron deficiency effect on somatodendritic compartment of cultured motoneurons

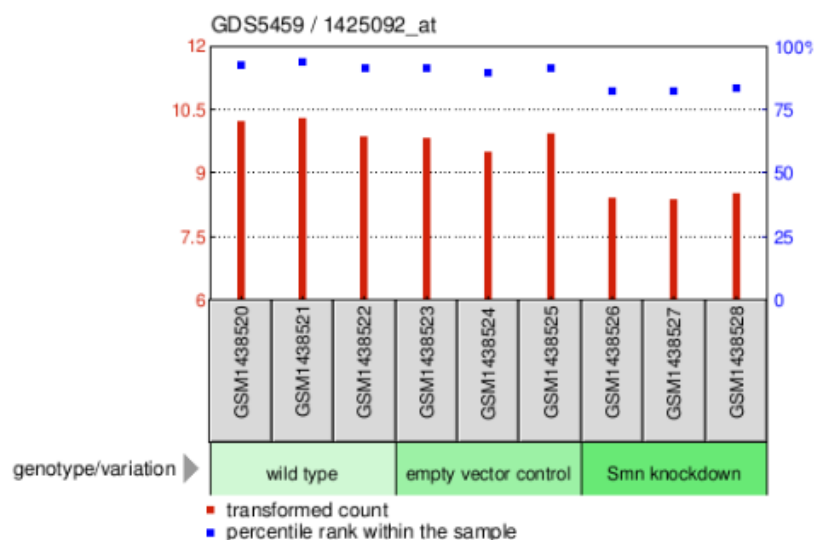
Annotation: Cdh10, cadherin 10

Organism: Mus musculus

Reporter: GPL1261, 1425092_at (ID_REF), GDS5459, 320873 (Gene ID), AF183946

DataSet type: Expression profiling by array, transformed count, 9 samples

ID: 126949598



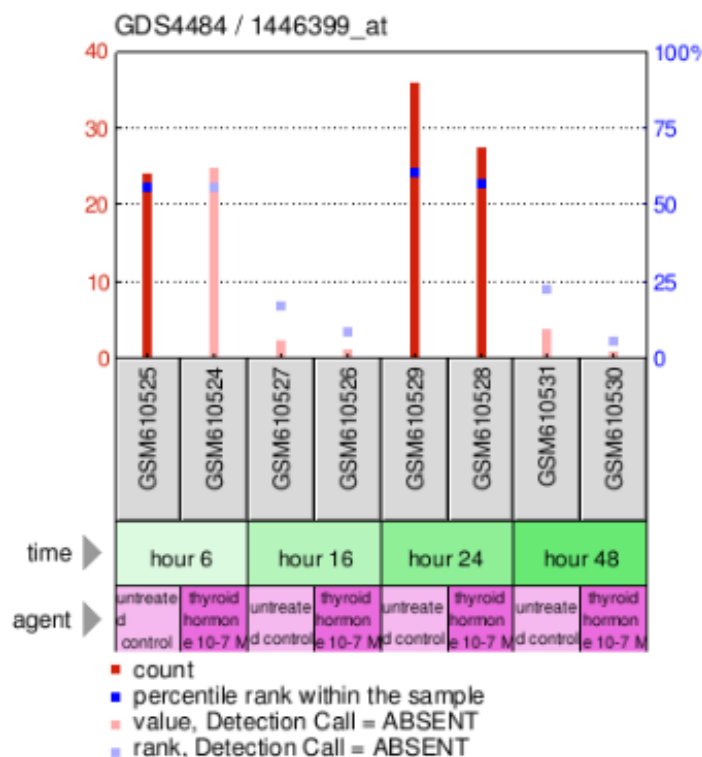
3. Cdh10 - Cerebellar neuronal cell response to thyroid hormone: time course

Annotation: Cdh10, cadherin 10

Organism: Mus musculus

Reporter: GPL1261, 1446399_at (ID_REF), GDS4484, 320873 (Gene ID), BB522094

DataSet type: Expression profiling by array, count, 8 samples
ID: 96756905



4. [CDH10 - Antiretroviral therapy effect on brain of patients with HIV-associated neurocognitive disorders](#)

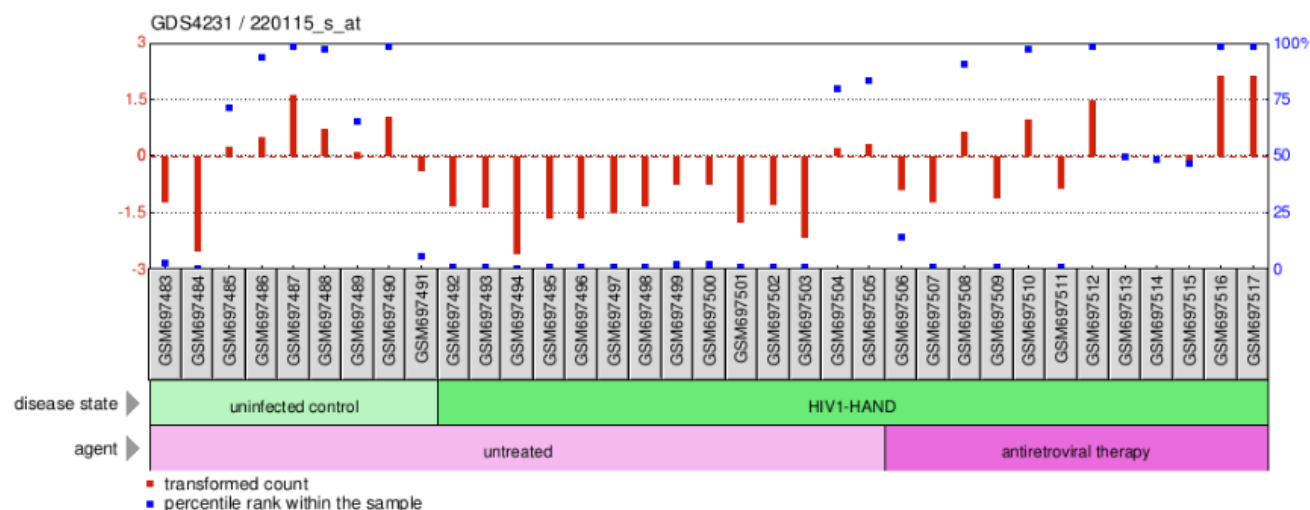
Annotation: CDH10, cadherin 10

Organism: Homo sapiens

Reporter: GPL570, 220115_s_at (ID_REF), GDS4231, 1008 (Gene ID), NM_006727

DataSet type: Expression profiling by array, transformed count, 35 samples

ID: 91055200



5. CDH10 - Chronic high-level alcohol consumption effect on brain: post-mortem hippocampus

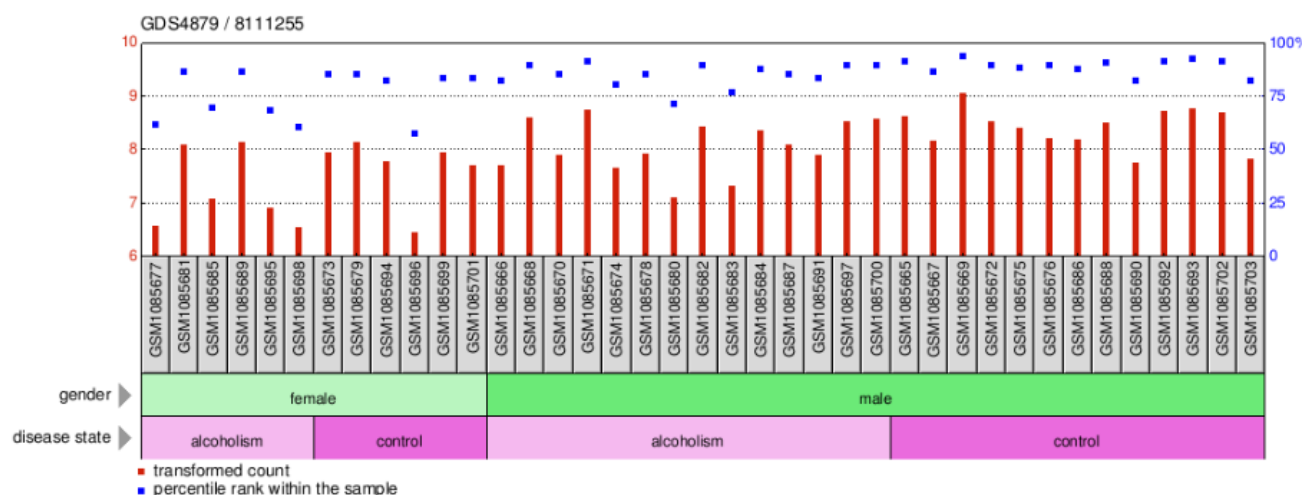
Annotation: CDH10, cadherin 10

Organism: Homo sapiens

Reporter: GPL6244, 8111255 (ID_REF), GDS4879, NM_006727, AB035303, AF039747, chr5:24487209-24645037 (SPOT ID)

DataSet type: Expression profiling by array, transformed count, 39 samples

ID: 106121937



6. CDH10 - Two types of HIV-associated neurocognitive impairment: brain regions

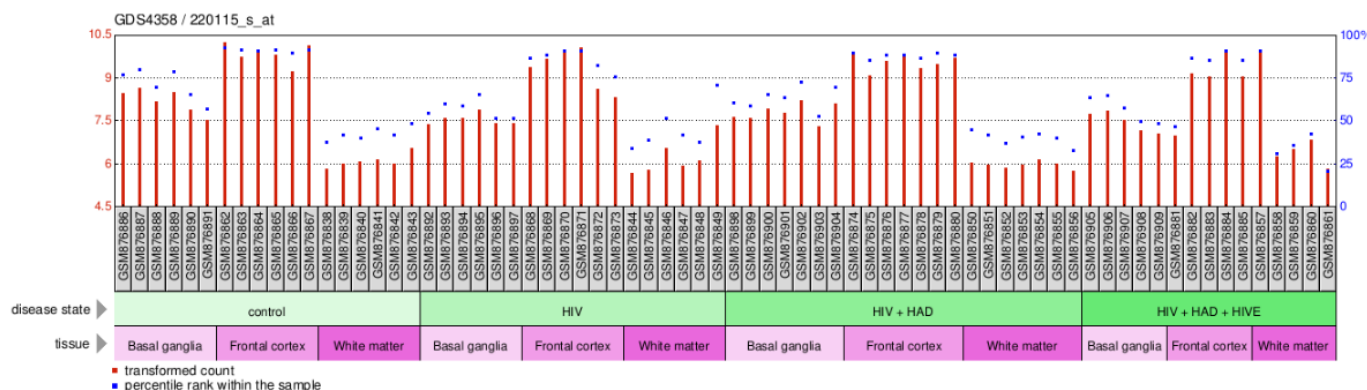
Annotation: CDH10, cadherin 10

Organism: Homo sapiens

Reporter: GPL570, 220115_s_at (ID_REF), GDS4358, 1008 (Gene ID), NM_006727

DataSet type: Expression profiling by array, transformed count, 72 samples

ID: 84179300



4. Transcription factors that drive the gene's expression

a. Harmonizome Integrated Knowledge About Genes & Proteins

Transcription factors (TFs) are proteins responsible for modulating gene transcription. The [Harmonizome](#) is a collection of processed datasets gathered to serve and mine knowledge about genes and proteins. Multiple datasets have been integrated, including transcription factor targets and transcription factor binding site profiles from the following resources:

ENCODE (Encyclopedia of DNA Elements), a compilation of functional elements in the human genome, including transcription factors and other circumstances under which a gene is active.

ChEA (ChIP Enrichment Analysis), a gene-set enrichment analysis tool tailored to test if query gene-sets are enriched with genes that are putative targets of transcription factors.

MotifMap, which uses databases of transcription factor binding motifs, refined genome alignments, and a comparative genomic statistical approach to provide comprehensive maps of candidate regulatory elements encoded in the genomes of model species

JASPAR, an open-access database of curated, non-redundant transcription factor-binding profiles stored as position frequency matrices (PFMs) for TFs across multiple species in six taxonomic groups.

CDH10 has 224 transcription factor associations. Selected associations are listed below.

Association	Dataset	Tissue/Cells	Organism	Genome Build
NANOG	CHEA Transcription Factor Binding Site Profiles	MESC	Mouse	
RCOR3	CHEA Transcription Factor Binding Site Profiles	MESC	Mouse	
JUN	CHEA Transcription Factor Binding Site Profiles	K562	Human	
SOX2	CHEA Transcription Factor Binding Site Profiles	LN229_GBM	Human	
OLIG2	CHEA Transcription Factor Targets			
ESR2	CHEA Transcription Factor Targets			
RUNX1	CHEA Transcription Factor Targets			
CREB1	CHEA Transcription Factor Targets			
YAP1	CHEA Transcription Factor Targets			
CHD1	ENCODE Transcription Factor Binding Site Profiles	H1-hESC		hg19_1
CEBPB	ENCODE Transcription Factor Binding Site Profiles	C2C12		mm9_1
H2AFZ	ENCODE Transcription Factor Binding Site Profiles	mammary epithelial cell		hg19_1
POLR2A	ENCODE Transcription Factor Binding Site Profiles	brain		mm9_1
POLR2A	ENCODE Transcription Factor Binding Site Profiles	MCF 10A		hg19_1

CDH10 has 11 predicted transcription factor associations.

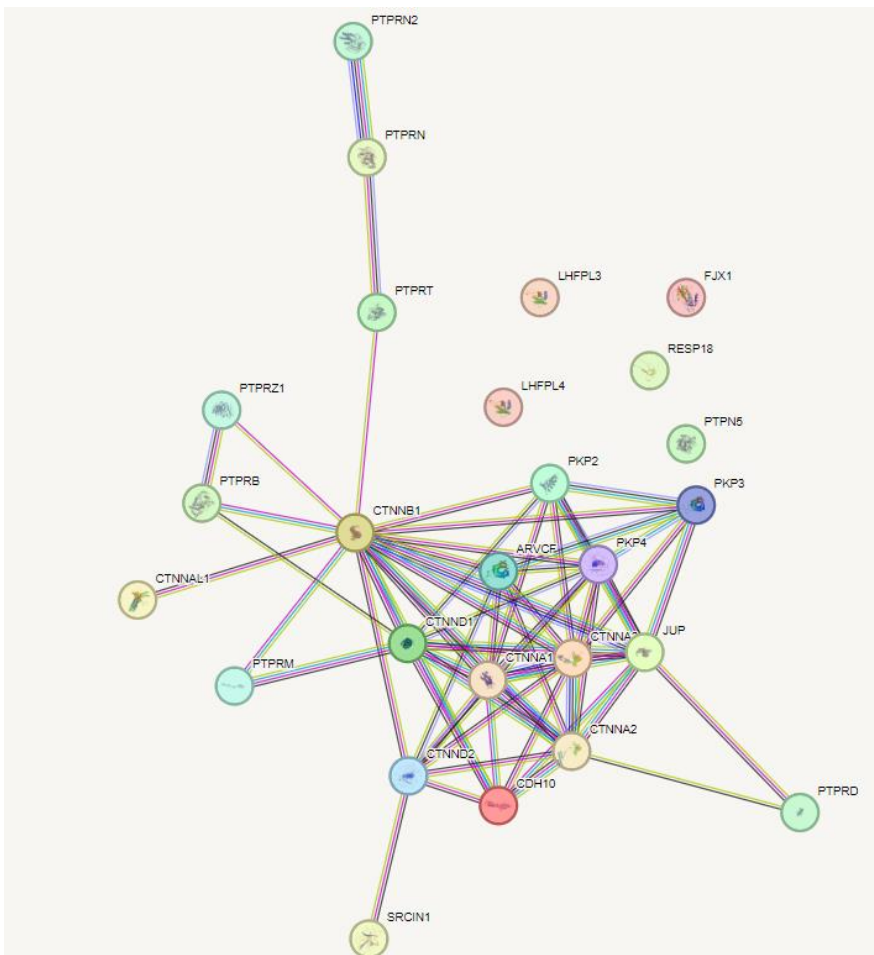
Association	Dataset
YY1	JASPAR Predicted Transcription Factor Targets
HINFP	JASPAR Predicted Transcription Factor Targets
GATA2	JASPAR Predicted Transcription Factor Targets
STAT3	JASPAR Predicted Transcription Factor Targets
FOXL1	JASPAR Predicted Transcription Factor Targets
PPARG	JASPAR Predicted Transcription Factor Targets
STAT4	MotifMap Predicted Transcription Factor Targets
TEF-1	MotifMap Predicted Transcription Factor Targets
POU5F1	MotifMap Predicted Transcription Factor Targets
YY1	MotifMap Predicted Transcription Factor Targets
MAFA	MotifMap Predicted Transcription Factor Targets

The download of all transcription factor associations is available as Supplementary Material (**CDH10_Harmonizome.xlsx**).

5. Protein-protein interaction (PPI) partners

a. STRING

STRING is a database of known and predicted protein-protein interactions (PPIs). The interactions include direct (physical) and indirect (functional) associations; they stem from computational prediction, from knowledge transfer between organisms, and from interactions aggregated from other (primary) databases. Data sources include genomic context predictions, high-throughput lab experiments, (conserved) co-expression, automated text-mining, and previous knowledge in databases. The download of all interactions is available as Supplementary Material (CDH10_STRING.xlsx)



Nodes:

Network nodes represent proteins

splice isoforms or post-translational modifications are collapsed, i.e. each node represents all the proteins produced by a single, protein-coding gene locus.

Node Color



*colored nodes:
query proteins and first shell of interactors*



*white nodes:
second shell of interactors*

Node Content



*empty nodes:
proteins of unknown 3D structure*



*filled nodes:
some 3D structure is known or predicted*

Edges:

Edges represent protein-protein associations

associations are meant to be specific and meaningful, i.e. proteins jointly contribute to a shared function; this does not necessarily mean they are physically binding each other.

Known Interactions



from curated databases



experimentally determined

Predicted Interactions



gene neighborhood



gene fusions



gene co-occurrence

Others



textmining



co-expression

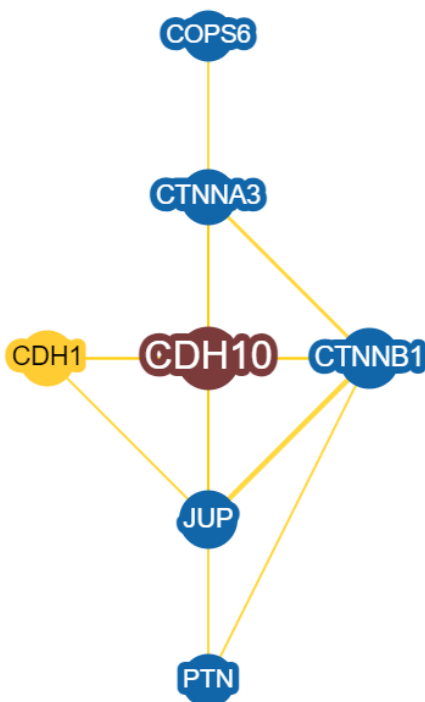


protein homology

b. BioGRID

BioGRID is an interaction repository with data compiled through comprehensive curation efforts. The download of all interactions is available as Supplementary Material ([CDH10_BIOGRID.xlsx](#)).

CDH10 has 6 unique interactions.

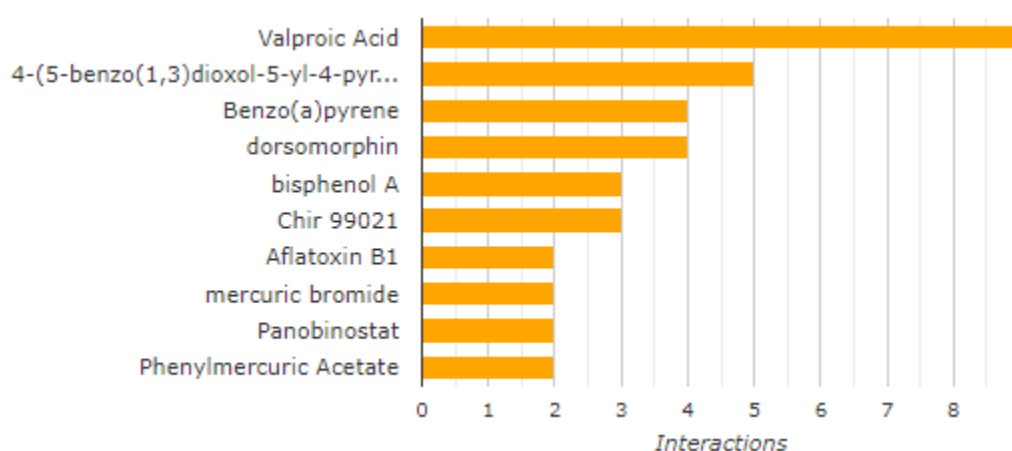


6. Information on which perturbations affect the gene

a. Comparative Toxicogenomics Database

[Comparative Toxicogenomics Database](#) (CTD) provides manually curated information about chemical–gene/protein interactions, chemical–disease and gene–disease relationships. The full list of interacting chemicals is provided as Supplementary Material (CDH10_CTD.xlsx).

Top Interacting Chemicals



7. Available chemical tools and their effects on the gene

a. Queried in the following databases to identify small molecule inhibitors

CDH10 gene was queried in the following databases to identify any known inhibitors or agonists and their effect(s) on the gene.

ChEMBL – No results

DrugBank – No results

PubMed –

The type-II cadherin CDH10 directly repressed by G9a through histone methylation. UNC0642, a specific small molecule inhibitor of G9a histone methyltransferase, rescued hypoxia-mediated gene repression of CDH10 in mice. [[Casciello et al., 2020](#)]

PubChem – No results

NCATS Inxight: Drugs – No results

b. Tractability assessments for small molecule

Small molecule tractability – [No results](#)

8. eQTLs that alter the gene expression in the brain tissue

a. GTEx eQTL Browser

An expression quantitative trait loci (eQTL) is a locus that explains a fraction of the genetic variance of a gene expression phenotype, and can be thought of as a 'molecular link' between genetic variation and organismal phenotypes. A search of the [GTEx](#) portal identified 1051 significant eQTLs for CDH10 gene for all tissues with 1 of them where significant in brain tissues. The full list of eQTLs is provided as Supplementary Material (CDH10_GTEx_eQTL.xlsx).

9. Known SNPs linked to function/disease

a. ClinVar

ClinVar is a freely accessible resource of archived and aggregated information that can be utilized to evaluate relationships between human sequence variation and phenotypes. A search in [ClinVar](#) produced 83 results, 46 of which are pathogenic. ClinVar reports a link of CDH10 gene with Neurodevelopmental disorder, Inborn genetic diseases, 5p partial monosomy syndrome, and Global developmental delay. The full list of results is available as Supplementary Material (CDH10_ClinVar.xlsx).

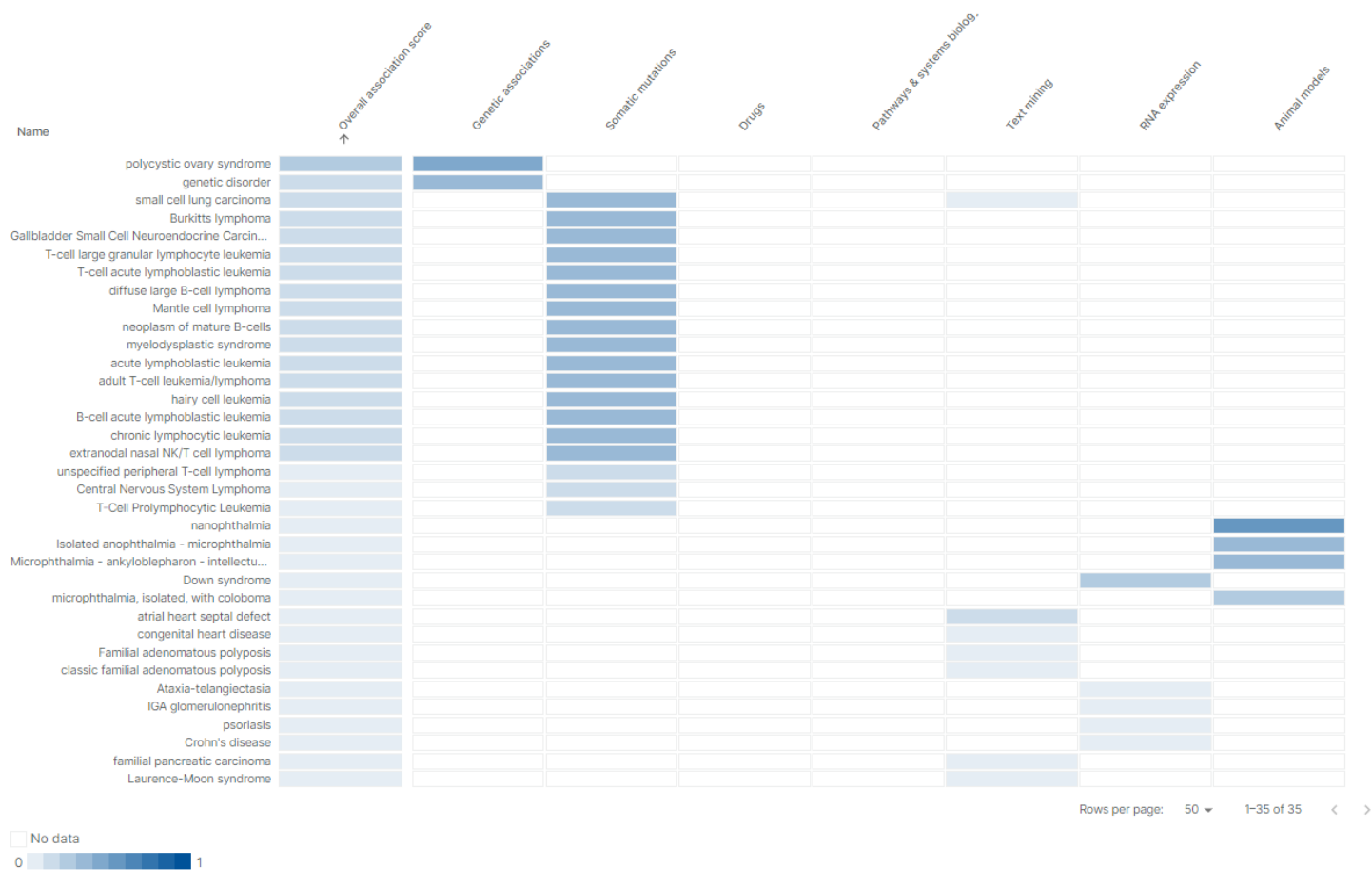
b. OMIM

OMIM (Online Mendelian Inheritance in Man) is a freely available resource that compiles information focused on the relationship between phenotype and genotype. [OMIM](#) describes a link of CDH10 gene with autism spectrum disorders (ASDs) and autosomal dominant high myopia.

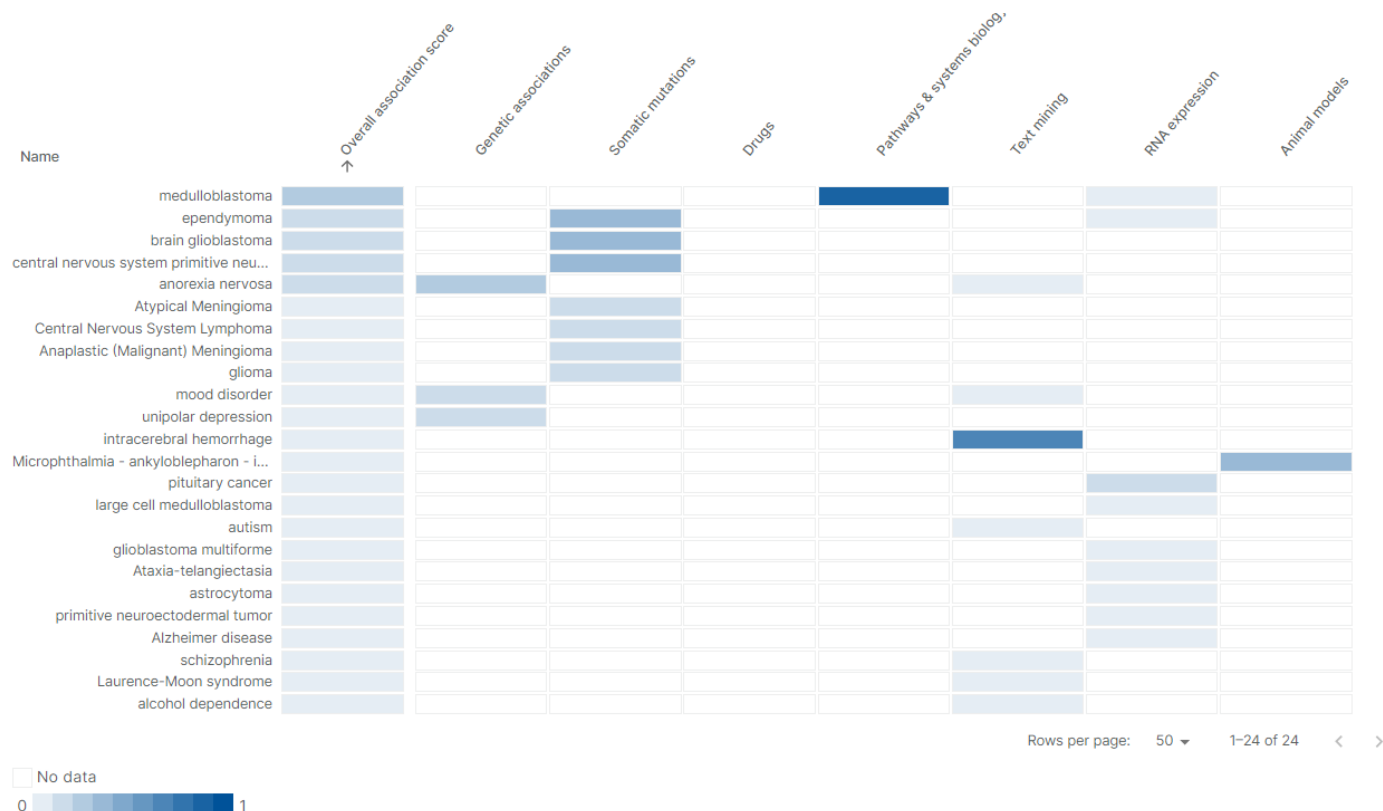
c. Open Targets

The Open Targets Platform is a resource that enables query and visualization of potential drug targets associated with disease, with the goal of helping to prioritize targets for further investigation. A search of CDH10 in [Open Target](#) revealed 169 disease associations. Genetic (n=35) and nervous system disease (n=24) associations are shown below.

Genetically linked with disease



Nervous system diseases



b. SNPs3D

SNPs3D is a website which assigns molecular functional effects of non-synonymous SNPs based on structure and sequence analysis. The corresponding data were generated by the John Moults Group ([Link](#)). A query of SNPs3D database found 4 SNPs related to the gene of interest ([SNP analysis on Disease Gene: CDH10 \(cadherin 10, type 2 \(T2-cadherin\)\)\)](#)).



refseq accession	snp	snp_id	svm profile	svm structure	molecular effect	model frequency role
NP_006718						
	V14G	35460631	-0.57 😞			
	V182I	34600303	-0.65 😞			
	S413F	1395027	-0.21 😞			
	V459G	13174039	-2.52 😞			

10. Major pathways

a. Pathways associated with the gene

Reactome pathways

- [Adherens junctions interactions](#)

Pathway Commons ([Link](#))

- [Wnt signaling pathway](#)
- [Cadherin signaling pathway](#)

11. Available knockouts of the gene and their phenotype

a. PubMed

Three type II cadherins, cadherins-6, 9, and 10, are selectively required for high-magnitude synaptic potentiation in the hippocampus. Presynaptic cadherin-9 binds postsynaptic cadherins-6 and 10 to regulate mushroom spine density and high-magnitude LTP in the CA1 SO layer. Cdh10 knockout mice have a significantly reduced density of mushroom spines specifically in the SO layer compared to wildtype mice. Spine analysis indicates Cdh6 and Cdh10 double knockout mice have a significantly more severe spine defect than loss of cadherin-10. CA1 neurons lacking both cadherins-6 and 10 have a 50% reduction of mushroom spines in the SO layer compared to wildtype. The more severe reduction in mushroom spines results in an overall reduction of spine head width in Cdh6 and Cdh10 double knockout SO spines. Loss of all three cadherins has no effect on the lower-magnitude LTP typically observed in the SR layer, demonstrating that cadherins-6, -9, and -10 are gatekeepers for high-magnitude LTP [[Basu et al., 2017](#)].

b. Mouse Genome Informatics (MGI)

The Mouse Genome Informatics (MGI) database is a comprehensive resource that integrates genetic, genomic, and biological data pertaining to the laboratory mouse. A search in [MGI](#) identified 8 targeted mutation and 3 endonuclease-mediated mutations for CDH10 (CDH10_MGI.xlsx).

Strain ID	Strain/Stock	Repository	State	Allele Symbol	Gene ID	Url
NM-CKO-205109	C57BL/6Smoc-Cdh10<em1(flox)Smoc>	SMOC	embryo	Cdh10<em1(flox)Smoc>	MGI:107436	link
NM-KO-2102372	C57BL/6Smoc-Cdh10<em1Smoc>	SMOC	embryo	Cdh10<em1Smoc>	MGI:107436	link

Additionally, 6 phenotypes from 1 allele in 1 genetic background have been reported for CDH10. The full list of results is available as Supplementary Material (CDH10_MGI.xlsx).

Allelic Composition	Genetic Background	Annotated Term	Phenotype Summary Category	Reference
Cdh10<tm2b(KOMP)Wtsi>/Cdh10<tm2b(KOMP)Wtsi>	C57BL/6N-Cdh10<tm2b(KOMP)Wt si>/Ucd	increased lean body mass	growth/size/body region	J:211773
Cdh10<tm2b(KOMP)Wtsi>/Cdh10<tm2b(KOMP)Wtsi>	C57BL/6N-Cdh10<tm2b(KOMP)Wt si>/Ucd	increased leukocyte cell number	hematopoietic system	
Cdh10<tm2b(KOMP)Wtsi>/Cdh10<tm2b(KOMP)Wtsi>	C57BL/6N-Cdh10<tm2b(KOMP)Wt si>/Ucd	increased lymphocyte cell number	immune system	
Cdh10<tm2b(KOMP)Wtsi>/Cdh10<tm2b(KOMP)Wtsi>	C57BL/6N-Cdh10<tm2b(KOMP)Wt si>/Ucd	increased monocyte cell number	hematopoietic system	
Cdh10<tm2b(KOMP)Wtsi>/Cdh10<tm2b(KOMP)Wtsi>	C57BL/6N-Cdh10<tm2b(KOMP)Wt si>/Ucd	increased neutrophil cell number	hematopoietic system	
Cdh10<tm2b(KOMP)Wtsi>/Cdh10<tm2b(KOMP)Wtsi>	C57BL/6N-Cdh10<tm2b(KOMP)Wt si>/Ucd	microphthalmia	vision/eye	

12. Major researchers working on the gene based on publications

Dr. Peter Clark

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Division of Investigative Sciences, Department of Histopathology, St Mary's Campus, Imperial College London, London, UK.

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[Pubmed link](#)

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Web page: https://bioscience.utah.edu/faculty/williams-megan/williams_megan.php

[Pubmed link](#)

13. Commercial antibodies

Antibody product name	Vendor name	Catalog number	Antibody description
Anti-cadherin 10 antibody [EPR5115] (ab134137)	abcam	ab134137	Rabbit monoclonal [EPR5115] to cadherin 10
Anti-cadherin 10 antibody (ab230309)	abcam	ab230309	Rabbit polyclonal to cadherin 10
Anti-cadherin 10 antibody [EPR5115] - BSA and Azide free (ab248697)	abcam	ab248697	Rabbit monoclonal [EPR5115] to cadherin 10 - BSA and Azide free
Anti-cadherin 10 antibody (ab196662)	abcam	ab196662	Rabbit polyclonal to cadherin 10
Cadherin-10 Antibody - BSA Free; Cadherin-10 Antibody (S07-2E2); Cadherin-10 Antibody	Novus Biologicals	NBP2-92600; NBP3-19842; NBP3-25341	Rabbit polyclonal to CDH10; Rabbit Monoclonal to CDH10; Rabbit polyclonal to CDH10
cadherin 10 (CDH10) (C-term) Rabbit Polyclonal Antibody;	OriGene	AP11486PU-N; AP11487PU-N	cadherin 10 (CDH10) (C-term) rabbit polyclonal antibody, Aff – Purified; cadherin 10 (CDH10) (N-

cadherin 10 (CDH10) (N-term) Rabbit Polyclonal Antibody			term) rabbit polyclonal antibody, Aff - Purified
CDH10 Antibody	CUSABIO	CSB-PA897318ESR1HU	Rabbit anti-Homo sapiens (Human) CDH10 Polyclonal antibody
CDH10 Antibody (PA5-119771); CDH10 Antibody (PA5-109503)	Invitrogen	PA5-119771; PA5-109503	Rabbit Polyclonal to CDH10
Anti-Cadherin 10 Rabbit Monoclonal Antibody	BOSTER	M10616	Anti-Cadherin 10 Rabbit Monoclonal Antibody
CDH10 Antibody	ProSci	13-498	Rabbit Polyclonal to CDH10

14. Information on Patents

Title	Patents	Applicant	Abstract/Summary
Use of brain-specific antigens to home, block and deliver cell-based treatments to the brain	WO2021173193A1	Wendell A. Lim, Milos Simic, Hideho Okada, Joseph H. CHOE, Payal B. WATCHMAKER	Provided herein is a cell comprising a recombinant nucleic acid encoding a transmembrane protein that has an extracellular binding domain that specifically binds to a brain-selective extracellular antigen, e.g., MOG, CDH10, PTPRZ1 or NRCAM, wherein the cell does not comprise a nucleic acid encoding an antigen-specific therapeutic that binds to a killing antigen expressed by a glioblastoma.
Breast cancer prediction, predicting of a progesteron receptor subtype and prediction of an antiprogesterin treatment on the basis of gene expression	RU2014147625A	Кэрол ЛАНГЕ, Тодд Филип КНАТСОН, Джейсон Басил НИКАС	1. A method for determining whether a malignant tumor contains cells expressing an active progesterone receptor (KR) and whether it will respond to therapeutic treatment with antiprogesterin, including: (a) obtaining a biological sample from a patient diagnosed with a malignant tumor; (b) determining in the cells of said biological sample at least one expression level of the sample of at least one gene selected from the group consisting of KBTBD11, RBPMS2, PLA2G48, FL112684, SH2D4B, RASCD2, CLDN8, and any combination thereof; and (c) comparing at least one expression level of the sample in step (b) with at least one expression level of said at least one gene in the cells of a wild-type (WT) control sample, wherein if the expression level of the sample is reduced compared to with the specified control sample, the malignant

			tumor responds to antiprogesterone treatment. 2. The method of claim 1, further comprising (d) determining the expression level in the cells of said biological sample of at least one gene selected from the group consisting of VCX, CHN2, AFAP1L2, PXMP4, THY1, ZNF26, CDH10, ZNF812 and any their combinations; and (e) comparing the expression level in step (d) with the expression levels of said at least one gene in a wild-type (WT) control sample, wherein the malignant tumor responds to antiprogesterone treatment if the expression level of said at least one gene is biological the sample is elevated compared to the specified control sample WT.3. The method according to any one of paragraphs. 1 or 2, in which the malignant tumor is selected from the group consisting of a tumor of the breast, ovary, endometrium, brain, lung, prostate gland, endoma.
Glioma molecular typing and prognosis markers	CN114854853A	刘春芳, 马展	The invention belongs to the field of biological and medical inspection, and relates to a group of tumor markers for glioma molecular typing, malignancy grading and prognosis evaluation and application thereof; the genes involved include: 30 genes such as ACVR1, BAX, BCL10, BMP8B, CANT1, CDH10, CDH12, CDK11B, CTSA, CXCR4, GDF15, IFITM3, ITGA1, ITGA3, ITGA4, ITGA11, ITGB1, ITGB1BP1, ITGB2, ITGB3BP, JUN, JUNB, LIF, NRG3, POU5F1, PRDM1, ROR2, STAT3, WNT5A, GDF15 and the like. The expression quantity of the gene (single or any combination) products (protein and RNA) is used for carrying out molecular typing, drug screening, malignancy classification, overall survival prediction of patients and the like on glioma. The related markers can be directly applied, and can also be combined with WHO pathological grading, 1p19q mutation state and the like.

15. NIH and ERC grants

a. [NIH RePORTER](#)

The NIH Research Portfolio Online Reporting Tools (RePORT) website provides access to reports, data, and analyses of NIH research activities, including information on NIH expenditures and the results of NIH-supported research. The RePORT Expenditures and Results (RePORTER) module

identifies 1 active CDH10-related grants funded by the NIH as shown below. A list of all CDH10-related grants is available as Supplementary Material (CDH10_NIH_RePORTER.xlsx).

Project Title	Project Number	Organization Name	Budget Start Date	Budget End Date	Funding IC(s)
Mechanisms Underlying Type II Cadherin Guided Assembly of Retinal Circuits	5R01EY030138-05	UNIVERSITY OF CALIFORNIA, SAN FRANCISCO	1/1/2024	12/31/2024	NEI

b. ERC

The [European Research Council \(ERC\)](#) is the premier European funding organization for excellent frontier research. It funds creative researchers of any nationality and age, to run projects based across Europe. It encourages the highest quality research in Europe through competitive funding and to support investigator-driven frontier research across all fields, based on scientific excellence. No CDH10-related grant funded by the ERC was found.

References

1. Basu R, Duan X, Taylor MR, Martin EA, Muralidhar S, Wang Y, Gangi-Wellman L, Das SC, Yamagata M, West PJ, Sanes JR, Williams ME. Heterophilic Type II Cadherins Are Required for High-Magnitude Synaptic Potentiation in the Hippocampus. *Neuron*. 2017 Sep 27;96(1):160-176.e8. doi: 10.1016/j.neuron.2017.09.009. Erratum in: *Neuron*. 2018 May 2;98 (3):658-668. PMID: [28957665](#); PMCID: PMC5634529.
2. Casciello F, Al-Ejeh F, Miranda M, Kelly G, Baxter E, Windloch K, Gannon F, Lee JS. G9a-mediated repression of CDH10 in hypoxia enhances breast tumour cell motility and associates with poor survival outcome. *Theranostics*. 2020 Mar 15;10(10):4515-4529. doi: 10.7150/thno.41453. PMID: [32292512](#); PMCID: PMC7150496.

Compiled List of Relevant Databases

1. GeneCards: <https://www.genecards.org/>
2. USCS Genome Browser: <https://genome.ucsc.edu/>

3. NCBI Structure Group: <https://www.ncbi.nlm.nih.gov/Structure/index.shtml>
4. The RCSB Protein Data Bank: <https://www.rcsb.org/>
5. AlphaFold Protein Structure Database: <https://www.alphafold.ebi.ac.uk/>
6. NCBI Annotation Pipeline: https://www.ncbi.nlm.nih.gov/genome/annotation_euk/process/
7. GTEx Portal: <https://gtexportal.org/home/>
8. The Human Brain Transcriptome (HBT) Project: <http://hbatlas.org/>
9. Allen Brain Map: <http://portal.brain-map.org/>
10. Barres Lab Brain RNA-Seq Portal: <http://www.brainrnaseq.org/>
11. DropViz: <http://dropviz.org/>
12. NCBI GEO Profiles: <https://www.ncbi.nlm.nih.gov/geoprofiles/>
13. Harmonizome: <https://maayanlab.cloud/Harmonizome/>
14. STRING: <https://string-db.org/>
15. BioGRID: <https://thebiogrid.org/>
16. Comparative Toxicogenomics Database: <http://ctdbase.org/>
17. ChEMBL: <https://www.ebi.ac.uk/chembl/>
18. DrugBank: <https://www.drugbank.ca/>
19. PubMed: <https://www.ncbi.nlm.nih.gov/pubmed/>
20. PubChem: <https://pubchem.ncbi.nlm.nih.gov/>
21. NCATS Inxight: <https://drugs.ncats.io/about>
22. NCBI ClinVar: <https://www.ncbi.nlm.nih.gov/clinvar/>
23. Online Mendelian Inheritance in Man (OMIM): <https://www.omim.org/>
24. Open Targets: <https://www.targetvalidation.org/>
25. Pathway Commons: <https://apps.pathwaycommons.org/>
26. Reactome: <https://reactome.org/>
27. Mouse Genomic Informatics (MGI): <http://www.informatics.jax.org/>
28. NIH Research Portfolio Online Reporting Tools (RePORT): <https://report.nih.gov/>
29. European Research Council (ERC): <https://erc.europa.eu/homepage>